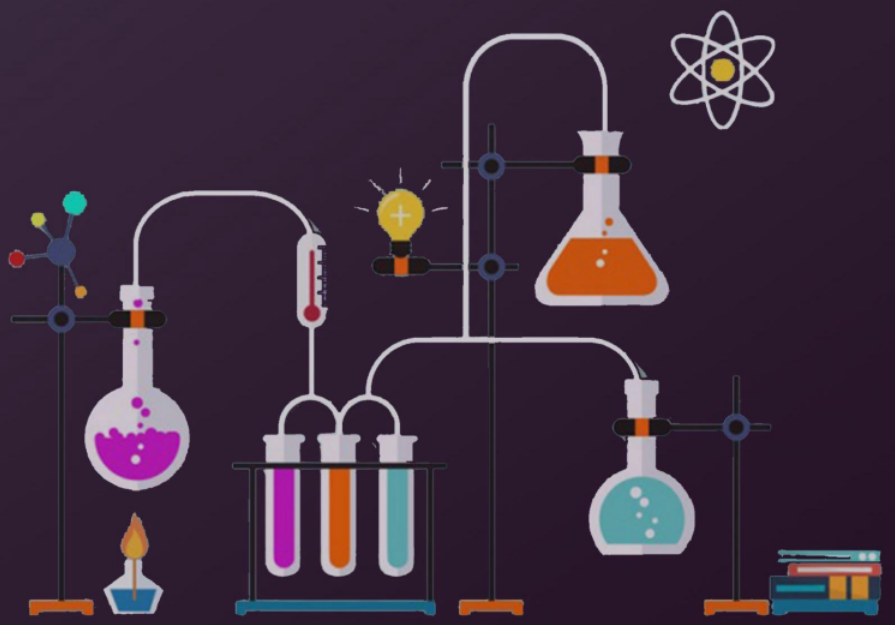
CHEMISTRY

Periodic Table



SHEET

eSaral.com

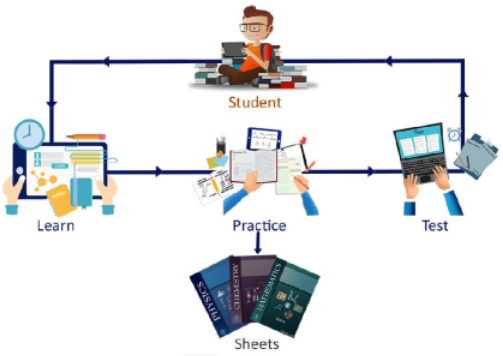
Contents

1.	Module Description	Page - ii
2.	Homework Index	Page - iii
3.	Exercise 1	Page - 1
4.	Exercise 1A	Page - 6
5.	Exercise 2	Page - 11
6.	Exercise 2A	Page - 15
7.	Answer Key	Page - 19

Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr.No.	Topic Name	Ex 1	Ex 1A	Ex 2	Ex 2A
1	Introduction to Periodic table	1-6	1, 2		
2	Group, Period and Blocks	7-16	31, 38, 39, 42, 44	1-6,8	6, 23, 25, 32
3	Periodicity,Effective Nuclear Charge and Atomic Radius	17-35	3, 25, 26	9,10	31
4	Atomic Radius in Detail		4, 20-22	4,5,11-20	1, 2, 28, 33, 34
5	Ionisation Energy	36-49	5, 7, 9, 17, 18, 19, 33, 43, 45	21,22,24,25	29
6	Applicatiopn of Ionisation Energy		6, 13, 15		10, 14, 15, 18-22, 35
7	Electron gain Enthalpy	50-56	7, 10-11, 14, 23, 24, 36, 41, 46	23,26-29	3, 5, 8, 11, 16-17, 24, 27, 30, 36
8	Electronegativity	57-66	12, 16, 27-30, 32, 34-35, 37, 40	30-32	4, 7, 9, 12, 13, 26
9	ACC			7,33	

Exercise - 1 Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Development of Periodic Table

Q1. Atomic wt. of P is 31 and Sb is 120. What will be the atomic wt. of As, as per Dobernier triad rule ?

(A) 151 (B) 75.5
(C) 89.5 (D) Unpredictable

Q2. Which of the following set of elements follow Newland's octave rule ?

(A) Be, Mg, Ca (B) Na, K, Rb
(C) F, Cl, Br (D) B, Al, Ga

Q3. Which are correctly matched ?

(a) Eka silicon – Be
(b) Eka aluminium – Ga
(c) Eka mangenese – Tc
(d) Eka scandium – B

(A) b, c (B) a, b, d
(C) a, d (D) All

Q4. The places that were left empty by Mendeleev were for ?

(A) Aluminium & Silicon
(B) Galium and germinium
(C) Arsenic and antimony
(D) Molybdenum and tungstun

Q5. The discovery of which of the following group of elements led to the failure of Rule of Octave?

(A) Inert gases
(B) Alkali metals
(C) Transuranic element
(D) Halogens

Q6. Atomic no. is the base of ?

(a) Lother meyer curve
(b) Newland octave rule
(c) Modern periodic table
(d) Doeberiener triad rule

(A) a (B) c
(C) d (D) b

Group, Period and Block

Q7. Which of the following statement is wrong?

(A) No inert gas is present in 7th period
(B) 3rd period contains 18 elements
(C) 1st period contains two non metals
(D) In p–block, metal, nonmetal and metalloids are present

Q8. From the list given below, elements which belong to the same group or sub-group are?

(A) Atomic number = 12, 20, 4, 88
(B) Atomic number = 8, 16, 34, 2
(C) Atomic number = 11, 18, 27, 5
(D) Atomic number = 24, 47, 42, 55

Q9. Which of the following set of elements belongs to same period ?

(A) Zn, Cd, Hg (B) Fr, Ra, U
(C) K, Ca, Ag (D) None

Q10. There are 10 neutrons in the nucleus of the element $_{Z}M^{19}$. It belongs to ?

(A) f-block (B) s-block
(C) d-block (D) p-block

Q11. Elements upto atomic no. 118 have been discovered till now. What will be the electronic configuration of the element possessing atomic no.108 ?

(A) $[Rn]5f^{14}\ 6d^{6}\ 7s^{2}$
(B) $6f^{14}\ 7d^{8}\ 7s^{2}$
(C) $[Rn]\ 5f^{14}\ 6d^{8}\ 7s^{0}$
(D) $[Xe]\ 4f^{14}\ 5d^{8}\ 6s^{2}$

Q12. For an element the quantum numbers of the last electron are : l = 3, m = – 2 and s = + 1/2. In which block of the periodic table is this element present ?

(A) s-Block (B) p-Block
(C) d-Block (D) f-Block

Q13. Elements with similar chemical properties?

(A) Occur only within the same period
(B) Have identical atomic mass (weight)
(C) Have identical number of neutrons
(D) Have the same number of electrons in the outer shell.

Q14. As applied to periodic table, which of the following sets include only magic numbers?

(A) 2, 8, 20, 28, 50, 82, 126
(B) 2, 8, 8, 18, 18, 32
(C) 2, 2, 8, 8, 18, 32
(D) 2, 8, 18, 18, 32, 32

Q15. An element recently discovered is Uub. Expected electronic configuration of the element is ?

(A) [Kr] $5s^2 4d^{10}$ (B) [Xe] $6s^2 4f^{14} 5d^8$
(C) [Rn] $7s^2 5f^{14} 6d^{10}$ (D) [Rn] $7s^2 5f^{10} d^0$

Q16. Tendency of I^-, Br^-, Cl^- and F^- to be oxidised is in order :

(A) $I^- > Br^- > Cl^- > F^-$
(B) $F^- > Cl^- > Br^- > I^-$
(C) $Cl^- > F^- > Br^- > I^-$
(D) $I^- = Br^- < Cl^- = F^-$

Z_{eff}, Screening Constant & Atomic Radius

Q17. The formula for effective nuclear charge is (if s is screening constant)

(A) Z - σ (B) Z + σ
(C) $Z\,\sigma^{-1}$ (D) Z σ

Q18. In sodium atom the screening is due to ?

(A) $3s^2$, $3p^6$ (B) $2s^1$
(C) $1s^2$, $2s^2$, $2p^6$ (D) $1s^2$, $2s^2$

Q19. The screening effect of d- electrons is ?

(A) Equal to the p - electrons
(B) Much more than p - electrons
(C) Same as f - electrons
(D) Less than p - electrons

Q20. Effective nuclear charge in group generally:

(A) Increases down the group
(B) Decreases down the group
(C) Remains constant
(D) First increases than decreases

Q21. Effective nuclear charge in a period generally :-

(A) Increases (B) Decreases
(C) Remains constant (D) First

Q22. Ionic radii are :-

(A) Directly proportional to square of effective nuclear charge
(B) Inversely proportional to effective nuclear charge
(C) Inversely proportional to square of effective nuclear charge
(D) Directly proportional to effective nuclear charge

Q23. Maximum size of first member of a period is due to ?

(A) Maximum number of shells
(B) Maximum screening effect
(C) Minimum Zeff
(D) All

Q24. Incorrect statement is ?

(A) Atomic size $\propto Z_{eff}$
(B) Atomic size $\propto$ negative charge
(C) Bond length $\propto Z_{eff}$
(D) Atomic size $\propto$ Screening effect

Q25. According to Slater's rule, order of effective nuclear charge for last electron in case of Li, Na and K is ?

(A) Li > Na > K (B) K > Na > Li

(C) Na > Li > K (D) Li < Na = K

Q26. Which of the following has largest radius ?

(A) $1s^2, 2s^2, 2p^6, 3s^2$

(B) $1s^2, 2s^2, 2p^6, 3s^2, 3p^1$

(C) $1s^2, 2s^2, 2p^6, 3s^2, 3p^3$

(D) $1s^2, 2s^2, 2p^6, 3s^2, 3p^5$

Q27. The ionic radii of N^{3-}, O^{2-} and F^- are respectively given by **?**

(A) 1.36, 1.40, 1.71

(B) 1.36, 1.71, 1.40

(C) 1.71, 1.40, 1.36

(D) 1.71, 1.36, 1.40

Q28. In which of the following compound, distance between two nuclei is maximum ?

(A) CsF (B) KI

(C) CsI (D) LiI

Q29. Highest size will be of ?

(A) Br^- (B) Br

(C) I^- (D) I

Q30. Which group of atoms have nearly same atomic radius ?

(A) Na, K, Rb, Cs (B) Li, Be, B, C

(C) Fe, Co, Ni (D) F, Cl, Br, I

Q31. Arrange the elements in increasing order of atomic radius Na, Rb, K, Mg ?

(A) Na < K < Mg < Rb

(B) K < Na < Mg < Rb

(C) Mg < Na < K < Rb

(D) Rb < K < Mg < Na

Q32. In which of the following pairs, both the species have nearly the same size ?

(A) Al, Ga (B) Li^+, Mg^{2+}

(C) Fe, Co (D) All the three

Q33. Identify the **wrong** statement in the following:

(A) Atomic radius of the elements increases as one moves down the first group of the periodic table.

(B) Atomic radius of the elements decreases as one-moves across from left to right in the 2nd period of the periodic table.

(C) Amongst isoelectronic species, smaller the positive charge on the cation, smaller is the ionic radius.

(D) Amongst isoelectronic species, greater the negative charge on the anion, larger is the ionic radius.

Q34. Moving left to right in the periodic table atomic density :-

(A) First increases and then decreases

(B) Remains the same

(C) Can't be said

(D) Generally increases

Q35. In a group moving top to down the atmoic density :-

(A) Decreases

(B) Remains the same

(C) Can't be said

(D) Generally increases

Ionisation Energy

Q36. The ionisation potential/energy of isotopes of an element will be ?

(A) Same

(B) Different

(C) Depends on atomic masses

(D) Depends on number of neutron

Q37. Compared to the first ionisation potential, the value of second ionisation potential of an element is ?

(A) Negligible (B) Smaller

(C) Greater (D) Double

Q38. Least ionisation energy will be of ?

(A) Be^{3+} (B) H

(C) Li^{+2} (D) He^{+}

Q39. Highest ionisation potential in a period is shown by ?

(A) Alkali metals

(B) Noble gases

(C) Halogens

(D) Alkali Earth Metals

Q40. In which of the following pairs, the ionisation energy of the first species is less than that of the second ?

(A) O^{-}, O^{2-} (B) S, P

(C) N, P (D) Be^{+}, Be

Q41. A sudden large jump between the values of 2nd and 3rd I.P. of an element would be associated with the electronic configuration-

(A) $1s^2, 2s^2\ 2p^6, 3s^1$

(B) $1s^2, 2s^2\ 2p^6, 3s^2\ 3p^5$

(C) $1s^2, 2s^2\ 2p^6, 3s^2\ 3p^2$

(D) $1s^2, 2s^2\ 2p^6\ 3s^2$

Q42. The second ionisation potentials in electron volts of oxygen and fluorine atoms are respectively given by ?

(A) 35.1, 38.3 (B) 38.3, 38.3

(C) 38.3, 35.1 (D) 35.1, 35.1

Q43. Correct orders of I^{st} I.P. are ?

(a) Li < B < Be < C

(b) O < N < F

(c) Be < N < Ne

(A) a, b (B) b, c

(C) a, c (D) a, b, c

Q44. Successive ionisation energies of an element 'X' are given below (in K. Cal)

IP_1	IP_2	IP_3	IP_4
165	195	556	595

Electronic configuration of the element 'X' is?

(A) $1s^2, 2s^2 2p^6, 3s^2 3p^2$

(B) $1s^2, 2s^1$

(C) $1s^2, 2s^2 2p^2$

(D) $1s^2, 2s^2 2p^6, 3s^{2\backslash}$

Q45. Ionisation energy increases in the order :-

(A) Be < B < C < N

(B) B < Be < C < N

(C) C < N < Be < B

(D) N < C < Be < B

Q46. Mg forms Mg(II) because of ?

(A) The oxidation state of Mg is + 2

(B) Difference between $I.P_1$ and $I.P_2$ is greater than 16.0 eV

(C) There are only two electrons in the outermost energy level of Mg

(D) Difference between $I.P_1$ and $I.P_2$ is less than 11 eV

Q47. (a) $M^{-}_{(g)} \rightarrow M_{(g)}$ (b) $M_{(g)} \rightarrow M^{+}_{(g)}$

(c) $M^{+}_{(g)} \rightarrow M^{+2}_{(g)}$ (d) $M^{+2}_{(g)} \rightarrow M^{+3}_{(g)}$

Minimum and maximum IP would be of ?

(A) a, d (B) b, c

(C) c, d (D) d, a

Q48. Which of the following electronic configuration belongs to least and most metallic character respectively ?

(a) $1s^2\ 2s^1$ (b) $5s^2\ 5p^5$

(c) $3s^2\ 3p^6\ 4s^1$ (d) $1s^2\ 2s^2\ 2p^5$

(A) a, b (B) d, c

(C) b, a (D) c, d

Q49. The correct values of ionization energies (in kJ mol^{-1}) of Be, Ne, He and N respectively are?

(A) 786, 1012, 999, 1256

(B) 1012, 786, 999, 1256

(C) 786, 1012, 1256, 999

(D) 786, 999, 1012, 1256

Electron Affinity

Q50. In the process $Cl_{(g)} + e^- \xrightarrow{\Delta H} Cl^-(g)$, ΔH is ?

(A) Positive (B) Negative

(C) Zero (D) None

Q51. In which case the energy released is minimum ?

(A) $Cl \rightarrow Cl^-$ (B) $P \rightarrow P^-$

(C) $N \rightarrow N^-$ (D) $C \rightarrow C^-$

Q52. Which of the following is energy releasing process ?

(A) $Cl^- \rightarrow Cl\,(g) + e^-$

(B) $O^-(g) + e^- \rightarrow O^{2-}$

(C) $O\,(g) \rightarrow O^+(g) + e^-$

(D) $O\,(g) + e^- \rightarrow O^-(g)$

Q53. Second electron affinity of an element is?

(A) Always exothermic

(B) Endothermic for few elements

(C) Exothermic for few elements

(D) Always endothermic

Q54. Which of the following configuration will have lowest electron affinity ?

(A) ns^2np^5 (B) ns^2np^2

(C) ns^2np^3 (D) ns^2np^4

Q55. The electron affinity value for the halogens shows the following trend :–

(A) Cl > F > Br > I

(B) F < Cl < Br < I

(C) F > Cl > Br > I

(D) Br > Cl > F > I

Q56. Which of the following element will form most stable bivalent anion?

(A) Fluorine (B) Oxygen

(C) Chlorine (D) Nitrogen

Electronegativity

Q57. Which of the following is affected by stable configuration of an atom :-

(a) Electronegativity

(b) Ionisation potential

(c) Electron affinity

Correct answer is :-

(A) Only electronegativity

(B) Only ionisation potential

(C) Electron affinity and ionisation potential

(D) All of the above

Q58. The correct set of decreasing order of electronegativity is ?

(A) Li, H, Na (B) Na, H, Li

(C) H, Li, Na (D) Li, Na, H

Q59. Outermost electronic configuration of the most electronegative element is ?

(A) ns^2np^3 (B) ns^2np^6

(C) ns^2 (D) ns^2np^5

Q60. On the Pauling's electronegativity scale, which element is next to F ?

(A) Cl (B) O

(C) Br (D) Ne

Q61. Which one of the following is incorrect ?

(A) An element which has high electro-negativity always has high electron gain enthalpy

(B) Electron gain enthalpy is the property of an isolated atom

(C) Electronegativity is the property of a bonded atom

(D) Both electronegativity and electron gain enthalpy are usually directly related to nuclear charge and inversely related to atomic size

Q62. Select the group where EN increases down the group :-

(A) Ca, Sr, Ba (B) Zn, Cd, Hg

(C) F, Cl, Br (D) Li, Na, K

Q63. Calculate the bond length of C–X bond if C – C bond length is 1.54 Å and X–X bond length is 1.2 Å and electronegativities of C and X are 2.0 and 3.0 respectively.

(A) 2.74 Å (B) 1.37 Å

(C) 1.46 Å (D) 1.28 Å

Q64. The increasing order of acidic nature of Li_2O, BeO, B_2O_3 is ?

(A) $Li_2O > BeO < B_2O_3$

(B) $Li_2O < BeO < B_2O_3$

(C) $Li_2O < BeO > B_2O_3$

(D) $Li_2O > BeO > B_2O_3$

Q65. The lowest electronegativity of the element from the following atomic number is ?

(A) 37 (B) 55

(C) 9 (D) 35

Q66. Select the correct order of acidic strength of oxides.

(A) $SO_2 > SO_3 > CO_2$

(B) $Cs_2O > K_2O > Na_2O$

(C) $CO_2 > N_2O_5 > B_2O_3$

(D) $SO_3 > CO_2 > B_2O_3$

Exercise - 1A *Concept Builder*

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. The statements that are true for the long form of the Periodic table are :

(A) it reflects the sequence of filling the electrons in the order of sub energy levels s, p, d & f

(B) it helps to predict the stable valency states of the elements

(C) it reflects trends in physical & chemical properties of elements

(D) it helps to predict the relative ionicity of the bond between any two elements.

Q2. Which of the following statements is/are true about the long form of modern periodic table ?

(A) Properties of the elements are the periodic function of their atomic number.

(B) There are 7 periods.

(C) There are 8 groups

(D) It has separate position for the isotopes.

Q3. Which of the following represent the correct order (s) of effective nuclear charge ?

(A) F > O > N > C (B) F < O < N < C

(C) Rb = K = Na > Li (D) I > Br > F > Cl

Q4. Which of the following represent the correct order (s) of the radii ?

(A) $I^+ < I < I^-$

(B) Atomic radii of Fe, Co, Ni are approximately equal

(C) O > N > F

(D) $Fe_3^+ < Fe_2^+ < Fe$

Q5. Considering the following ionisation steps:

$A(g) \rightarrow A^+(g) + e^-$ $\Delta H = 100$ eV

$A(g) \rightarrow A^{2+}(g) + e^-$ $\Delta H = 250$ eV

Select the correct statements :

(A) IE_1 of A(g) is 100 eV

(B) IE_1 of $A^+(g)$ is 150 eV

(C) IE_2 of A(g) is 150 eV

(D) IE_2 of A(g) is 250 eV

Q6. Which of the following are correct ?

(A) $IE_2(Mg) < IE_2$ (Na)

(B) IE (N) > IE (P)

(C) Atomic size Mg^{+2} >Atomic size(Mg^+)

(D) IE of Na < Mg < Al

Q7. Which of the following represent the correct order of ionization energies ?

(A) F > O > N > C (B) F > N > O > C

(C) F > Cl > Br > I (D) I > Br > F > Cl

Q8. Which of the following represent the correct order of electron affinity ?

(A) F > O > C (B) Na > K > Rb

(C) Cl > Br > I (D) I > Br > F > Cl

Q9. Process $Na^+ \xrightarrow{I} Na_{(g)} \xrightarrow{II} Na_{(s)}$

(A) In (I) energy released

(B) In (II) energy is absorbed

(C) In (II) energy is released

(D) In (I) energy absorbed

Q10. Which of the following ions is most unlikely to exist ?

(A) Li^- (B) F^-

(C) B^- (D) Be^-

Q11. The electron affinity

(A) Of carbon is greater than oxygen

(B) Of fluorine is less than iodine

(C) Of Cl^- is less than Cl

(D) Of S is more than oxygen

Q12. Amongst the following statements, which is/ are correct?

(A) Electronegativity of sulphur is greater than that of oxygen.

(B) Electronegativity of oxygen is larger than that of sulphur.

(C) Electron gain enthalpy of fluorine is most negative

(D) Electron gain enthalpy of chlorine is most negative

Q13. Which of the following statement is/are not correct?

(A) I.P. increases down the group

(B) IP of s–block elements is less than corresponding d– block elements

(C) If ΔIP >16 eV higher oxidation state is more stable

(D) IP of halogen elements is maximum in their respective period

Q14. Select equations having endothermic step -

(A) $S^-(g) \rightarrow S^{2-}(g)$

(B) $Na^+(g) + Cl^-(g) \rightarrow NaCl(s)$

(C) $N(g) \rightarrow N^-(g)$

(D) $Al^{2+}(g) \rightarrow Al^{3+}(g)$

Q15. Diagonal relationship is shown by ?

(A) Be and Al (B) Li and Mg

(C) Mg and Al (D) B and P

Q16. Two elements M and X (both from same period) form oxide M_2O and X_2O. If X_2O is acidic and M_2O is basic then select CORRECT option(s) about M and X :

(A) M > X (size)

(B) (IE) X > M

(C) (EN) M > X

(D) (Acidic strength) MOH > XOH

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for (Ques. 17 to 19)

The IE1 and IEII in kJ/mol of a few elements are given in the following table :

Element	IE_I in kJ/Mol	IE_{II} in kJ//mol
P	2372	5251
Q	520	7300
R	900	1760
S	1680	3380

Q17. Which of the above element is likely to be alkali metal ?

(A) P (B) Q

(C) R (D) S

Q18. Which of the above element is likely to be alkaline earth metal ?

(A) P (B) Q

(C) R (D) S

Q19. Which of the above element is likely to be noble gas ?

(A) P (B) Q

(C) R (D) S

Paragraph for (Ques. 20 to 22)

A, B and C are hydroxy-compounds of the elements X, Y and Z respectively. X, Y and Z are in the same period of the periodic tabe. A gives an aqueous solution of pH less than seven. B reacts with both strong acids and strong alkalis. C gives an aqueous solution which is strongly alkaline.

Q20. Which of the following statements is/are true?

I. The three elements are metals.

II. The electronegativities decrease form X to Y to Z.

III. The atomic radius decreases in the order X, Y and Z

IV. X, Y and Z could be phosphorus, aluminium and sodium respectively.

(A) I, II, III only correct

(B) I, III only correct

(C) III, IV only correct

(D) II, III, IV only correct

Q21. Find out correct order of acedic strength in water.

(A) HF > HCl > HBr > HI

(B) HCl > HF > HBr > HI

(C) HFO > HClO > HBrO > HIO

(D) HF < HCl < HBr < HI

Q22. In the following hydroxy compounds which are present in paragraph. Find out maximum and minimum electronegativity of X, Y and Z.

(A) Z and X

(B) X and Y

(C) Y and Z

(D) All elements having same E.N.

Paragraph for (Ques. 23 to 24)

First electron gain enthalpy (in $\frac{kJ}{mol}$) of few elements are given below :

Elements	ΔH_{eg}
I	–60
II	–45
III	–328
IV	–295
V	+ 48

Answer the following questions on the basis of above data:

Q23. Which element may be an inert gas ?

(A) I (B) III (C) IV (D) V

Q24. Which element is most non-metallic among all the elements ?

(A) I (B) II (C) III (D) IV

Paragraph for (Ques. 25 to 27)

According to I. C. slater effective nuclear charge , Z^*, due to screening , is not exactly equal to the actual nuclear charge Z of the nucleus of the atom. Z^* depends on the type of orbital in which the electron is housed , and on the ability of other electrons in more penetrating orbitals to screen the electron in question from the nucleus.

The relative extent to which the various orbitals penetrate the electron clouds of other orbitals is s > p > d > f. The effective nuclear charge Z^* due to screening is given by , $Z^* = Z - \sigma$. Where Z is the atomic number and σ is the slater screening constant values : $0 < \sigma < Z$.

Q25. What is the value of Z^* for Rb (37) ?

(A) 1.7 (B) 1.3

(C) 2.2 (D) 34.8

Q26. What is Z^* for last electron of Cu (29) ?

(A) 29 (B) 24.8

(C) 7.85 (D) 4.2

Q27. In which of the following cases the concept of Z^* is applicable ?

(A) Ionisation energy increases as Z^* increases

(B) Electronegativity of the atom increases as Z^* increases

(C) Atomic size decreases along the period as Z^* increases

(D) All of the above

Paragraph for (Ques. 28 to 30)

The real meaning of the word periodic in any classification of elements is that when the elements are arranged in order of increasing atomic numbers in the same period or group there is a gradual change (i.e. increase or decrease) in a particular pattern. In a period the gradual change of electronic configuration from member to member. Some of the properties which depends upon electronic configuration of elements such as screening effect, atomic radii, ionization potential, electron affinity etc. follow the general trend of periodicity.

Q28. The correct order of ionic radii of the species N_3^-, O_2^-, Na^+ & F^- is ?

(A) $Na^+ < F^- < O_2^- < N_3^-$

(B) $F^- < O_2^- < N_3^- < Na^+$

(C) $O_2^- < N_3^- < F^- < Na^+$

(D) $N_3^- < Na^+ < F^- < O_2^-$

Q29. The electronegativities C,N, Si & P are in the order of –

(A) P < Si < C < N

(B) Si < P < N < C

(C) Si < P < C < N

(D) P < Si < N < C

Q30. The sizes of given species are such that :

(A) I– > I > I+ (B) I > I– > I+

(C) I > I+ > I– (D) I+ > I– > I

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q31. Match Column–I (atomic number of elements) withColumn–II (position of element in periodic table) and select the correct answer using the codes given below :

Column-I	**Column-II**
(A) 19	(P) p-block
(B) 22	(Q) f-block
(C) 32	(R) d-block
(D) 64	(S) s-block

Q32.

Column-A		Column-B	
(A)	Vertical column in periodic table	(i)	IE_1
(B)	$A(g) + \text{Energy} \rightarrow A^+(g) + e^-$	(ii)	Period
(C)	Horizontal row in periodic table	(iii)	EA_2
(D)	$A(g) + e^- \rightarrow A^-(g) + \text{Energy}$	(iv)	Group
(E)	$A^+(g) + \text{Energy} \rightarrow A^{2+}(g) + e^-$	(v)	EA_1
(F)	$A^-(g) + e^- \rightarrow A^{2-}(g) + \text{Energy}$	(vi)	IE_2

Q33.

Column I	**Column-II**
Configuration	Ionisation energy in k J/ mol
(A) ns^2	(P) 2100
(B) ns^2np^1	(Q) 1400
(C) ns^2np^3	(R) 800
(D) $ns^2 np^6$	(S) 900

Q34.

Column-I	**Column-II**
(A) Increasing ionisation potential	(P) Cl > S > P
(B) Increasing electronegativity	(Q) N < O < F
(C) Decreasing Z_{eff}	(R) O < N < F
(D) Decreasing electron affinity	(S) O > C > N

Q35. Match the column :

Column-I	**Column-II**
(A) Cl	(P) Metal
(B) F	(Q) Highest electron gain enthalpy
(C) Cu	(R) Most Electronegative element
(D) He	(S) Highest ionisation energy.

Q36. I^{st} electron gain enthalpy in kJ/mol

Column-I	**Column-II**
(A) 66	(P) Cl
(B) –333	(Q) F
(C) 99	(R) Be
(D) –348	(S) Ne

Q37.

Column-I	**Column-II**
(A) Increasing atomic size	(P) Cl < O < F
(B) Decreasing atomic radius	(Q) Li < Be < B
(C) Increasing electronegativity	(R) Si < Al < Mg
(D) Increasing effective nuclear charge	(S) N > O > F

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q38. Find out the atomic number of element whose IUPAC name is Unnilpentium.

Q39. A particular atom(not ion) having atomic number between 22 to 30 has magnetic moment equal to 1.73 B.M. Then find the atomic number of the element which is next to it in the periodic table.

Q40. Select total number of acidic compounds out of given below :

$CsOH$, $OC(OH)_2$, $SO_2(OH)_2$, $Sr(OH)_2$, $Ca(OH)_2$, $Ba(OH)_2$, $BrOH$, $NaOH$, O_2NOH

Q41. Total number of enthalpy(s) (out of given eight) of A(g) which is/are **not** associated with conversion of $A_{(g)}^{-} \rightarrow A_{(g)}^{+4}$:

IE_1, IE_2, IE_3, IE_4, IE_5, IE_6, EGE_1, EGE_2

(IE = ionization energy, EGE = electron gain enthalpy)

Q42. How many element belong to same group Na, Mg, K, Rb, N, O, C, Cs

Q43. Upto argon find the number of elements which have lower IE_1 as compared to He.

Q44. Element with atomic number 114, which well known element would you expect it to resemble? Write it's atomic number (between Z=1 to 10).

SUBJECTIVE TYPE

Q45. The 1st ionisation energy of potassium is less than that of Cu but the reverse is true for 2nd ionisation energy.

Q46. Of the two elements given in each of the following sets choose the element having a more negative electron gain enthalpy.

(A) N and O (B) F and Cl

Exercise - 2 *Brain Booster*

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. The number of d- electrons in Mn^{2+} is equal to that of :-

(A) p-electrons in N

(B) s-electron in Na

(C) d-electrons in Fe^{+2}

(D) p-electrons in O^{-2}

Q2. Among the following, which species is/are paramagnetic ?

(i) Sr^{2+} (ii) Fe^{3+}

(iii) Co^{2+} (iv) S^{2-}

(v) Pb^{2+}

(A) i, iv, v (B) i, ii, iii

(C) ii, iii (D) iv, v

Q3. The correct order of stability of Al^{+}, Al^{+2}, Al^{+3} is ?

(A) $Al^{+3} > Al^{+2} > Al^{+}$

(B) $Al^{+2} > Al^{+3} > Al^{+}$

(C) $Al^{+2} < Al^{+} > Al^{+3}$

(D) $Al^{+3} > Al^{+} > Al^{+2}$

Q4. A transition element X has the configuration [Ar] $3d^4$ in its +3 oxidation state. Its atomic number is ?

(A) 25 (B) 26

(C) 22 (D) 19

Q5. Which of the following lanthanoid ions is diamagnetic?

(Atoms,Ce = 58, Sm = 62, Yb =70)

(A) Yb^{2+} (B) Ce^{2+}

(C) Sm^{2+} (D) Eu^{2+}

Q6. If Aufbau and Hund's rule are not used, then Incorrect statement is :

(A) K^+ would be coloured ion

(B) Na will be in same s-block (if these rules are true)

(C) Cu would be s-block element

(D) magnetic moment of Cr (24) would-be zero

Q7. Which of the following arrangements shows the correct order of decreasing paramagnetism ?

(A) N > Al > O > Ca

(B) N > O > Al > Ca

(C) O > N > Al > Ca

(D) O > N > Ca > Al

Q8. Total number of p block elements among given atomic number 83, 79, 42, 64, 37, 54, 34 are?

(A) 3 (B) 4

(C) 5 (D) 6

Q9. Rank the 4p, 4d and 4f orbitals of increasing degree to which the electrons present in them are shielded by inner electrons ?

(A) 4d > 4f > 4p

(B) 4f > 4d > 4p

(C) 4p > 4d > 4f

(D) 4d > 4p > 4f

Q10. Which of the following option is correct ? [State T for true and F for false]

(i) On moving from left to right in a period screening constant value increases

(ii) In s, p, d, f-orbitals the order of screening effect is s > p > d > f in a given shell

(iii) In any period from left to right effective nuclear charge increases so electronegativity also increases

(iv) When σ increases, Z_{eff} also increase

(A) TTTF (B) FFFT

(C) FFTT (D) TTFF

Q11. Which of the following order is incorrect against the property indicated ?

(A) Mg < Ar < Na (2nd I.E.)

(B) Be < F < Cl ($\Delta H_{e.g.}$)

(C) Rb < Na < K < Ca (atomic radius)

(D) P < S < N (electronegativity)

Q12. Arrange in the increasing order of atomic radii of the following elements F, O, C, Cl, Br :-

(A) F < O < C < Cl < Br

(B) F < C < O < Cl < Br

(C) F < Cl < Br < O < C

(D) C < O < F < Cl < Br

Q13. These are 3 elements A, B and C.Their atomic number are Z_1, Z_2, Z_3 respectively.

If $Z_3 - Z_1 = 2$ and $\frac{Z_1 + Z_3}{2} = Z_2$ and the electronic configuration of element C is [Ar] $3d^2 4s^2$ then correct order of atomic radius is ?

(A) $A^{+2} < B^{+3} < C^{+4}$ (B) $A^{+2} = B^{+3} = C^{+4}$

(C) $A^{+2} > B^{+3} > C^{+4}$ (D) $B^{+3} < A^{+2} = C^{+4}$

Q14. The radii of F, F^-,O and O^{2-} are in the order of ?

(A) $O^{2-} > F^- > F > O$

(B) $F^- > O^{2-} > F > O$

(C) $O^{2-} > O > F^- > F$

(D) $O^{2-} > F^- > O > F$

Q15. Arrange the following in order of increasing atomic radii Na, Si, Al, Ar :-

(A) Na < Si < Al < Ar

(B) Si < Al < Na < Ar

(C) Ar < Al < Si < Na

(D) Na < Al < Si < Ar

Q16. Mg, Mg^{2+}, Al and Al^{3+} are arranged in decreasing order of size 1 > 2 > 3 > 4. Species which are present at 1 and 4 position respectively are :

(A) Al, Mg^{2+} (B) Mg, Al^{3+}

(C) Mg^{2+}, Al (D) Mg, Al^{3+}

Q17. Reason of lanthanoid contraction is ?

(A) Negligible screening effect of 'f' orbitals

(B) Increasing nuclear charge

(C) Decreasing nuclear charge

(D) Decreasing screening effect

Q18. For the element (X), student (A) measured its readius as 102 nm, student (B) as 103 nm and (C) as 100 nm using same apparatus. Their teacher explained that measurements were correct by saying that recorded values by (A), (B) and (C) are :

(A) crystal, van der Waals and covalent radii

(B) covalent, crystal and van der Waals radii

(C) van der Waals, ionic and covalent

(D) none is correct

Q19. Largest in size out of Na^+, Ne and F^- is :

(A) Na^+

(B) Ne

(C) F^-

(D) all are equal

Q20. The single covalent radius of P is 0.11 nm. The single covalent radius of Cl will be :

(A) smaller than P

(B) greater than P

(C) same as P

(D) twice of P

Q21. Amongst the following, the incorrect statement is ?

(A) IE_1 (Al) < IE_1 (Mg)

(B) IE_1 (Na) < IE_1(Mg)

(C) IE_2 (Mg) > IE_2 (Na)

(D) IE_3 (Mg) > IE_3 (Al)

Q22. Four successive members of the first row transition elements are listed below with their atomic numbers.Which one of them is expected to have the highest third ionization enthalpy ?

(A) Vanadium (Z = 23)

(B) Manganese (Z = 25)

(C) Chromium (Z = 24)

(D) Iron (Z = 26)

Q23. Electronegativity and electron affinity of an element A are X and Y respectively. Hence, ionisation potential of A is :

(A) $\frac{X+Y}{2}$ (B) 2X – Y

(C) 2Y – Y (D) 2X + Y

Q24. Correct order of I.E. in Ni, Pd and Pt

(A) Pt > Pd > Ni (B) Pt > Ni > Pd

(C) Ni > Pd > Pt (D) Pd > Pt > Ni

Q25. Following the transition elements, $(IE)_1$ drops abruptly in Ga, In and Tl. This is due to :

(A) decrease in effective nuclear charge

(B) increases in atomic radius

(C) removal of an electron from the singly occupied np orbitals of higher energy than the ns-orbitals of Zn, Cd and Hg.

(D) none is correct

Q26. The second electron gain enthalpies (in kJ mol^{-1}) of oxygen and sulphur respectively are :

(A) – 780, + 590 (B) – 590, +780

(C) + 590, + 780 (D) + 780, + 590

Q27. Increasing order of Electron affinity for following configuration.

(a) $1s^2, 2s^2\ 2p^2$

(b) $1s^2, 2s^2\ 2p^4$

(c) $1s^2, 2s^2\ 2p^6\ 3s^2\ 3p^4$

(d) $1s^2, 2s^2\ 2p^6, 3s^2\ 3p^3$

(A) d < a < b < c (B) d < a < c < b

(C) a < b < c < d (D) a < b < d < c

Q28. Screening effect is not observed in :

(A) He

(B) Li

(C) Be

(D*) in all cases

Q29. Select correct statement :

(A) Electron affinity of nitrogen is much lower than that of its neighbouring elements carbon and oxygen

(B) Electron affinity of F is higher than that of chlorine

(C) Both are correct

(D) None is correct

Q30. Electronegativity is the measurement of capacity of an atom by which :

(A) Electrons get replled

(B) Electrons get attracted

(C) Gain of electron

(D) Loss of proton

Q31. (A), (B), (C) are elements in the third short period. Oxide of (A) is ionic, that of (B) is amphoteric and of (C) a giant molecule. (A), (B) and (C) will have atomic number in the order :

(A) (A) < (B) < (C)

(B) (C) < (B) < (A)

(C) (A) < (C) < (B)

(D) (B) < (A) < (C)

Q32. Which of the following oxides is not expected to react with sodium hydroxide ?

(A) BeO (B) B_2O_3

(C) CaO (D) SiO_2

Q33. The ground state electronic configurations of the elements U, V, W, X, and Y (these symbols do not have any chemical significance) are as follows

U $1s^2 2s^2 2p^3$

V $1s^2 2s^2 2p^6 3s^1$

W $1s^2 2s^2 2p^6 3s^2 3p^2$

X $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5 4s^2$

Y $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6$

Determine which sequence of elements satisfy the following statements :

(i) Element forms a carbonate which is not decomposed by heating

(ii) Element is most likely to form coloured ionic compounds

(iii) Element has largest atomic radius

(iv) Element forms only acidic oxide

(A) V W Y U

(B) V X Y W

(C) V W Y X

(D) V X W U

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Select correct order of size :

(A) $Ti^{2+} < Ti < Zr$ (B) $Ti^{2+} < Ti < Hf$

(C) $Zr^{2+} < Zr \approx Hf$ (D) $Hf^{2+} < Hf \approx Zr$

Q2. Which of the following orders of atomic / Ionic radius is correct ?

(A) $B < Al \approx Ga$ (B) $Sc > Cu < Zn$

(C) $C < O < N$ (D) $Al^{+3} < Al^{+2} < Al^{+}$

Q3. Which of the following orders for electron affinity is /are correct ?

(A) $S > O < Se$ (B) $Cl > F$

(C) $N > P$ (D) $C > N$

Q4. AB exists as A^{+} B^{-} if :-

(A) $(IP)_A < (IP)_B$

(B) $(EA)_A < (EA)_B$

(C) $(EN)_A < (EN)_B$

(D) Size of A < size of B

Q5. Select the correct statement(s).

(A) The value of electron gain enthalpy of an element can be -ve or +ve.

(B) In the periodic table, metallic character of the elements increases down the group and decreases sacross the period.

(C) The Cl^{-} & S^{2-} are isoelectronic species but first one is not smaller in size than the second.

(D) Ionization enthalpy of an atom is equal to electron gain enthalpy of cation.

Q6. If Aufbau's principle and Hund's rule were not followed.

(A) K would have been a d–block element & a paramagnetic substance

(B) Cu would have been a s–block element.

(C) Cr would have been a diamagnetic substance

(D) None of these

Q7. Which of the following show amphoteric behaviour ?

(A) $Zn(OH)_2$ (B) BeO

(C) Al_2O_3 (D) $Pb(OH)_2$

Q8. Electron affinify of the elements or ions shown correct are ?

(A) S > O (B) $O > S^{-}$

(C) $O^{-} > S^{-}$ (D) $N^{-} > S$

Q9. In halogens, which of the following properties increases from iodine to fluroine ?

(A) Ionisation energy

(B) Electronegativity

(C) Bond length

(D) Electron affinity

Q10. In which of the following set of elements 1st element is more metallic than second ?

(A) Ba, Ca (B) Sb, Sn

(C) Ge, S (D) Na, F

Q11. Which of the following order is correct ?

(A) P < Si < Be < Mg < Na (Metallic character)

(B) $Mg^{+2} < Na^{+} < F^{-} < O^{2-}$ (Ionic radius)

(C) Li < B < Be < C < N < O (2nd ionization energy)

(D) Cl < F < S < O (electron gain enthalpy)

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for (Ques. 12 to 13)

Nature of bond can be predicted on the basis of electronegativity of bonded atoms, greater difference in electronegativity (X), more will be the polarity of bond, and polar bond are easily broken in polar solvent like water. For hydroxy acids $X_O - X_A$ difference predicts the nature of oxide formed by the element A.

$|X_O - X_A| > |X_O - X_H|$ then A–O–H shows basic nature (NaOH)

$|X_O - X_A| < |X_O - X_H|$ then A–O–H shows acidic nature (H–O–Cl)

With the help of EN values [$EN_A = 1.8$, $EN_B = 2.6$, $EN_C = 1.6$, $EN_D = 2.8$] answer the following questions for the compounds HAO, HBO, HCO, HDO.

Q12. Compounds whose aqueous solution is acidic and order of their acidic strength is ?

(A) AOH, COH; AOH < COH

(B) HDO, HBO; HDO > HBO

(C) AOH, COH; AOH > COH

(D) HDO, HBO; HDO < HBO

Q13. Compounds whose aqueous solution is basic and order of their basic strength is ?

(A) AOH, COH; AOH < COH

(B) HDO, HBO ; HDO > HBO

(C) AOH, COH ;AOH > COH

(D) HDO, HBO ; HDO < HBO

Paragraph for (Ques. 14 to 17)

Four elements P, Q, R & S have ground state electronic configuration as:

P → $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^3$

Q → $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^1$

R → $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^2\ 4p^3$

S → $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^2\ 4p^1$

Q14. Comment which of the following option represent the correct order of true (T) & false (F) statement.

I size of P < size of Q

II size of R < size of S

III size of P < size of R (appreciable difference)

IV size of Q < size of S (appreciable difference)

(A) TTTT (B) TTTF

(C) FFTT (D) TTFF

Q15. Order of IE_1 values among the following is?

(A) P > R > S ≈ Q (B) P < R < S < Q

(C) R > S > P > Q (D) P > S > R > Q

Q16. Considering the elements F, Cl, O and S correct order of their electron affinity value is -

(A) F > Cl > O > S (B) F > O > Cl > S

(C) Cl > F > S > O (D) O > F > S > Cl

Q17. Which process involves maximum release of energy ?

(A) $O_{(g)} + e^- \rightarrow O^-_{(g)}$ (B) $S_{(g)} + e^- \rightarrow S^-_{(g)}$

(C) $F_{(g)} + e^- \rightarrow F^-_{(g)}$ (D) $Cl_{(g)} + e^- \rightarrow Cl^-_{(g)}$

Paragraph for (Ques. 18 to 23)

Ionization energies of five elements in kcal/mol are given below :

Atom	I	II	III
P	300	549	920
Q	99	734	1100
R	118	1091	1652
S	176	347	1848
T	497	947	1500

Q18. Which element is a noble gas ?

(A) P (B) T

(C) R (D) S

Q19. Which element form stable unipositive ion?

(A) P (B) Q

(C) R (D) S

Q20. The element having most stable oxidation state +2 is ?

(A) Q (B) R

(C) S (D) T

Q21. Which is a non-metal (excluding noble gas)?

(A) P (B) Q

(C) R (D) S

Q22. If Q reacts with fluorine and oxygen, the molecular formula of fluoride and oxide will be respectively ?

(A) QF_3, Q_2O_3 (B) QF, Q_2O

(C) QF_2, QO (D) None of these

Q23. Which of the following pair represents elements of same group ?

(A) Q, R (B) P, Q

(C) P, S (D) Q, S

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q24. Match the characteristics mentioned in Column II with the process in Column I.

Column-I	**Column-II**
(A) $O(g) + e^- \rightarrow O^-(g)$	(P) Positve electron gain enthalpy
(B) $O^-(g) + e^- \rightarrow O^{2-}(g)$	(Q) Negative electron gain enthalpy
(C) $Na^-(g) \rightarrow Na(g) + e^-$	(R) Exothermic
(D) $Mg^+(g) + e^- \rightarrow Mg(g)$	(S) Endothermic

Q25.

Column-I **(Consider elements of 4th period)**	**Column-II**
(A) Number of elements which have ns^1 configuration	(P) 4
(B) n + l for last electron of last element	(Q) 5
(C) Maximum unpaired electron in any of the element	(R) 6
(D) (n + l) for outermost electron of 1st element	(S) 3

Q26.

Column-I	**Column-II** **(Increasing order)**
(A) $Fe < Fe^{2+} < Fe^{3+}$	(P) Electronegativity
(B) O < Te < Se < S	(Q) Ionization energy
(C) $Mg^{2+} < Na^+ < F^- < N^{-3}$	(R) Electron affinity
(D) C < N < O < F	(S) Radius
	(T) Acidic character of hydride of element

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q27. The number of pairs, in which EA of the second element is more than that of the first element is ?

[O, S], [C, N], [O, N], [N, P],

[Cl^+, F^+], [K^+, Na^+]

Q28. Size of H^- is smaller than how many elements among these ?

H, Li, Be, B, C, N, O, F, F^-

Q29. The number of elements from the following group having I.E. greater than O (Z = 8)

Li, Be, B, C, N,

F, Ne, S, Si

Q30. Among F, O, S, Al, Zn, He and N the number of elements having electron affinity less than Cl is/are?

Q31. Find the Z_{eff} value on the last electron entered into Sc.

SUBJECTIVE TYPE

Q32. Use the following system of naming elements in which first alphabets of the digits are written collectively,

0	1	2	3	4
nil	uni	bi	tri	quad
5	6	7	8	9
pent	hex	sept	oct	enn

to write three-letter symbols for the elements with atomic number 101 to 109.

Q33. Arrange the following ions in increasing order of their radius ?

V^{+5}, K^+, S^{2-}, Mn^{+7}, Ca^{+2}, Cl^-, P^{3-}

Q34. The oxidation number that iron does not exhibit in its common compounds or in its elemental state is-

Q35. The first ionisation energy of beryllium is greater than that of lithium but reverse is true for the second ionisation energy.

Q36. How many chlorine atoms will be ionised ($Cl \rightarrow Cl^+ + e^-$) by the energy released from the process $Cl + e^- \rightarrow Cl^-$ for 6.023×10^{23} atom (IP for Cl = 1250 kj $mole^{-1}$ and EA = 350 KJ $mole^{-1}$)

Answer Key

Ex-1									
1	2	3	4	5	6	7	8	9	10
B	A	A	B	A	B	B	A	B	D
11	12	13	14	15	16	17	18	19	20
A	D	D	B	C	A	A	C	D	C
21	22	23	24	25	26	27	28	29	30
A	B	C	A	D	A	C	C	C	C
31	32	33	34	35	36	37	38	39	40
C	D	C	A	D	A	C	B	B	B
41	42	43	44	45	46	47	48	49	50
D	C	D	D	B	D	A	B	C	B
51	52	53	54	55	56	57	58	59	60
C	D	D	C	A	B	C	C	D	B
61	62	63	64	65	66				
A	B	D	B	B	D				

Ex-1A

1. ABCD **2.** AB **3.** AC **4.** ABD **5.** ABC

6. AB **7.** BC **8.** ABC **9.** AC **10.** D

11. CD **12.** BD **13.** ACD **14.** A,C,D **15.** A,B

16. A,B **17.** B **18.** C **19.** A **20.** C

21. D **22.** A **23.** D **24.** C **25.** C

26. C **27.** D **28.** A **29.** C **30.** A

31. A → S, B → R, C → P, D → Q

32. (A) → IV, (B) → I, (C) → II, (D) → V (E) → VI (F) → III

33. (A) → S, (B) → R, (C) → Q, (D) → P

34. (A) → R, (B) → Q, (C) → P (D) → S

35. A → Q, B → R, C → P, D → S

36. A → R, B → Q, C → S, D → P

37. (A) → R (B) → S (C) → P (D) → Q

38. 105 **39.** 30 **40.** 4 ($OC(OH)_2$, $SO_2(OH)_2$, BrOH, O_2NOH)

41. 3 (EGE_2IE_5, IE_6,) **42.** 4 (Na, K, Rb,Cs) **43.** 017

44. 6 **46.** (A)-O (B)-Cl

Ex-2									
1	2	3	4	5	6	7	8	9	10
B	C	D	A	A	A	B	C	B	A
11	12	13	14	15	16	17	18	19	20
C	A	C	D	B	B	A	A	B	B
21	22	23	24	25	26	27	28	29	30
C	B	B	A	C	D	A	C	A	B
31	32	33							
A	C	B							

Ex-2A

1. A,B,C,D **2.** A,B,D **3.** A,B **4.** A,B,C **5.** A,B,D

6. A,B,C **7.** A,B,C,D **8.** A,B **9.** A,B **10.** A,C,D

11. A,B **12.** B **13.** A **14.** B **15.** A

16. C **17.** D **18.** B **19.** B,C **20.** C

21. A **22.** B **23.** A

24. (A) $\rightarrow$ Q,R, (B) $\rightarrow$ P,S, (C) $\rightarrow$ S, (D) $\rightarrow$ Q,R

25. A $\rightarrow$ S, B$\rightarrow$ Q, C $\rightarrow$ R, D $\rightarrow$ P

26. (A) $\rightarrow$ P,Q,R, (B) $\rightarrow$ R, (C) $\rightarrow$ S, (D) $\rightarrow$ P,T

27. 4 **28.** 0 **29.** 3 **30.** 7 **31.** 3

32. 101-Unu, 102-Unb, 103-Unt, 104-Unq, 105-Unp, 106-Unh, 107-Uns, 108-Uno, 109-Une

33. $Mn^{+7} < V^{+5} < Ca^{+2} < K^{+} < Cl^{-} < S^{2-} < P^{3-}$

34. 4 **36.** 1.686×10^{23}